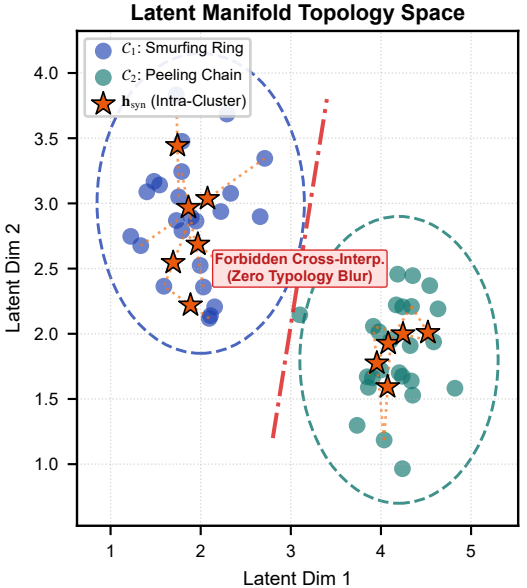
Three-Step Typology-Clustered Latent GraphSMOTE & Bilinear Edge Synthesis Pipeline



Step 1: Latent Typology Partitioning

- Minority illicit nodes clustered into $C_1, \ldots, C_K$
- Cosine similarity threshold $\tau_{\text{sim}} \geq 0.75$
- Preserves distinct criminal topological signatures

Step 2: Intra-Cluster Latent Interpolation

- Sample seed pair $u, v \in C_k$ within same cluster
- Convex combination: $\mathbf{h}_{\text{syn}} = (1-\rho)\mathbf{h}_u + \rho\mathbf{h}_v$
- Generates realistic synthetic minority representations

Step 3: Bilinear Edge Gen. & Flow Invariants

- Localized bilinear link head: $\hat{\mathbf{A}}_{\text{syn},\,w} = \sigma(\mathbf{h}_{\text{syn}}^T \mathbf{W}_{\text{edge}} \mathbf{h}_w)$
- Linear transaction amount interp: preserves $\Phi_{\text{flow}} \approx 1.0$
- 100% quarantined to $\mathcal{D}_{\text{train}}$: zero lookahead or test leakage